PID Controller via Pole Placement

(通过极点配置的PID控制器设计)

PID Controller via Pole Placement

- **Proper PID Controller Structure** (正则PID控制器结构)

$$C(s)=K_P+\frac{K_I}{s}+\frac{K_D s}{\tau_D s+1}$$

$$=K_P\left(1+\frac{1}{T_r s}+\frac{T_d\, s}{\tau_D s+1}\right)$$

$$=\frac{\left(K_D+K_P\tau_D\right)s^2+\left(K_P+K_I\tau_D\right)s+K_I}{\tau_D s^2+s}$$

$T_r=\dfrac{K_P}{K_I}$, Reset time (复位时间)

$T_d=\dfrac{K_D}{K_P}$, Derivative time (微分时间)

- **Equivalent Controller Form** (等效控制器形式)

$$C(s)=\frac{b_{c2}s^2+b_{c1}s+b_{c0}}{s^2+a_{c1}s}$$

or

$$b_{c2}=\frac{K_D+K_p\tau_D}{\tau_D},\quad b_{c0}=\frac{K_I}{\tau_D}$$

$$b_{c1}=\frac{K_p+K_I\tau_D}{\tau_D},\quad a_{c1}=\frac{1}{\tau_D}$$

$$\tau_D=\frac{1}{a_{c1}},\quad K_p=\left(b_{c1}-K_I\right)\tau_D$$

$$K_I=b_{c0}\tau_D,\quad K_D=\left(b_{c2}-K_p\right)\tau_D$$

PID Controller via Pole Placement

- **Determine PID Gains via Pole Placement (通过极点配置确定PID增益)**

$$D_{pn}(s)\left(s^2+a_{c1}s\right)+N_{pn}(s)\left(b_{c2}s^2+b_{c1}s+b_{c0}\right)=A_{Cd}(s)$$

Case 1: τ_D is pre-determined based on noise attenuation consideration (考虑噪声衰减预先确定τ_D)

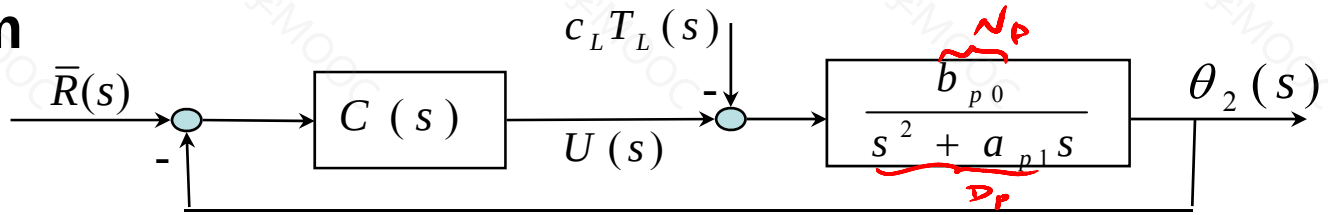
Only the Controller parameters b_{c2},b_{c1},b_{c0},or k_p,k_D and k_I are free to chosen. Thus, can only place three CL poles at the desired locations. So for plant with n>1,cannot arbitrarily place all CL poles. (只有控制器参数b_{c2},b_{c1},b_{c0}或 k_p,k_D 和 k_I可以自由选择。因此，只能配置三个闭环极点到期望的位置。所以对于n>1的系统，不能任意配置所有的闭环极点)

Case 2: τ_D is automatically determined from pole placement design (根据极点配置设计自动确定τ_D)

Four controller parameters to choose, can place four CL poles at wish. So for plant with $n\leq 2$, we can arbitrarily place all CL poles. (有四个控制器参数可选择，可任意配置四个闭环极点。对于$n\leq 2$的系统，我们可以任意配置所有的闭环极点)

PID Control of Rigid Belt ECP Emulator

- **System**



$b_{p0} = 623$
$a_{p1} = 3.5$

Would like to have magnitude of real parts of all CL poles larger than 20 for fast enough response if possible. **(希望所有闭环极点的实部都大于20，以便响应尽可能地快)**

Case 1: τ_D is pre-determined (预先确定τ_D)

$$\frac{1}{\tau_D} \geq 10 \times 20 \Rightarrow \tau_D \leq \frac{1}{200} = 0.005,$$

$$\text{let } \tau_D = 0.005, \Rightarrow a_{c1} = 200$$

$$A_{CL}(s) = (s^2 + a_{p1}s)(s + a_{c1})s + b_{p0}(b_{c2}s^2 + b_{c1}s + b_{c0})$$

$$= s^4 + (a_{c1} + a_{p1})s^3 + (a_{p1}a_{c1} + b_{p0}b_{c2})s^2 + b_{p0}b_{c1}s + b_{p0}b_{c0}$$

As n_L=2+2=4, but we only have three controller parameters to choose, so let us choose **(由于n_L=2+2=4，但我们只有三个控制器参数去选择，所以选择)**

$$p_{1,2d} = -20 \pm 20j, p_{3d} = -40, p_{4,d} = -\alpha$$

PID Control of Rigid Belt ECP Emulator

$$
\begin{aligned}
A_{CLd}(s) &= (s^2 + a_{p1}s)(s + a_{c1})s + b_{p0}(b_{c2}s^2 + b_{c1}s + b_{c0}) \\
&= s^4 + (a_{c1} + a_{p1})s^3 + (a_{p1}a_{c1} + b_{p0}b_{c2})s^2 + b_{p0}b_{c1}s + b_{p0}b_{c0} \\
&= s^4 + 203.5s^3 + (700 + 623b_{c2})s^2 + 623b_{c1}s + 623b_{c0} \\
&= (s + 20 - 20j)(s + 20 + 20j)(s + 40)(s + \alpha)
\end{aligned}
$$

$$
\Rightarrow \begin{cases} \alpha = 123.5 \\ b_{c2} = 18.6 \\ b_{c1} = 527 \\ b_{c0} = 6340 \end{cases}
$$

PID Control of Rigid Belt ECP Emulator

Case 2: τ_D is automatically determined (τ_D自动确定)

Since we can arbitrarily place all CL poles, let us choose (既然我们可以任意配置所有的CL极点，让我们选择)： $p_{1,2d} = -20 \pm 20j, p_{3,4d} = -40$

$$\Rightarrow A_{CLd}(s) = (s-(-20+20j))(s-(-20-20j))(s-(-40))(s-(-40))$$
$$= [(s+20)^2 + 20^2](s+40)^2$$
$$= s^4 + 120s^3 + 5600s^2 + 128000s + 1280000$$

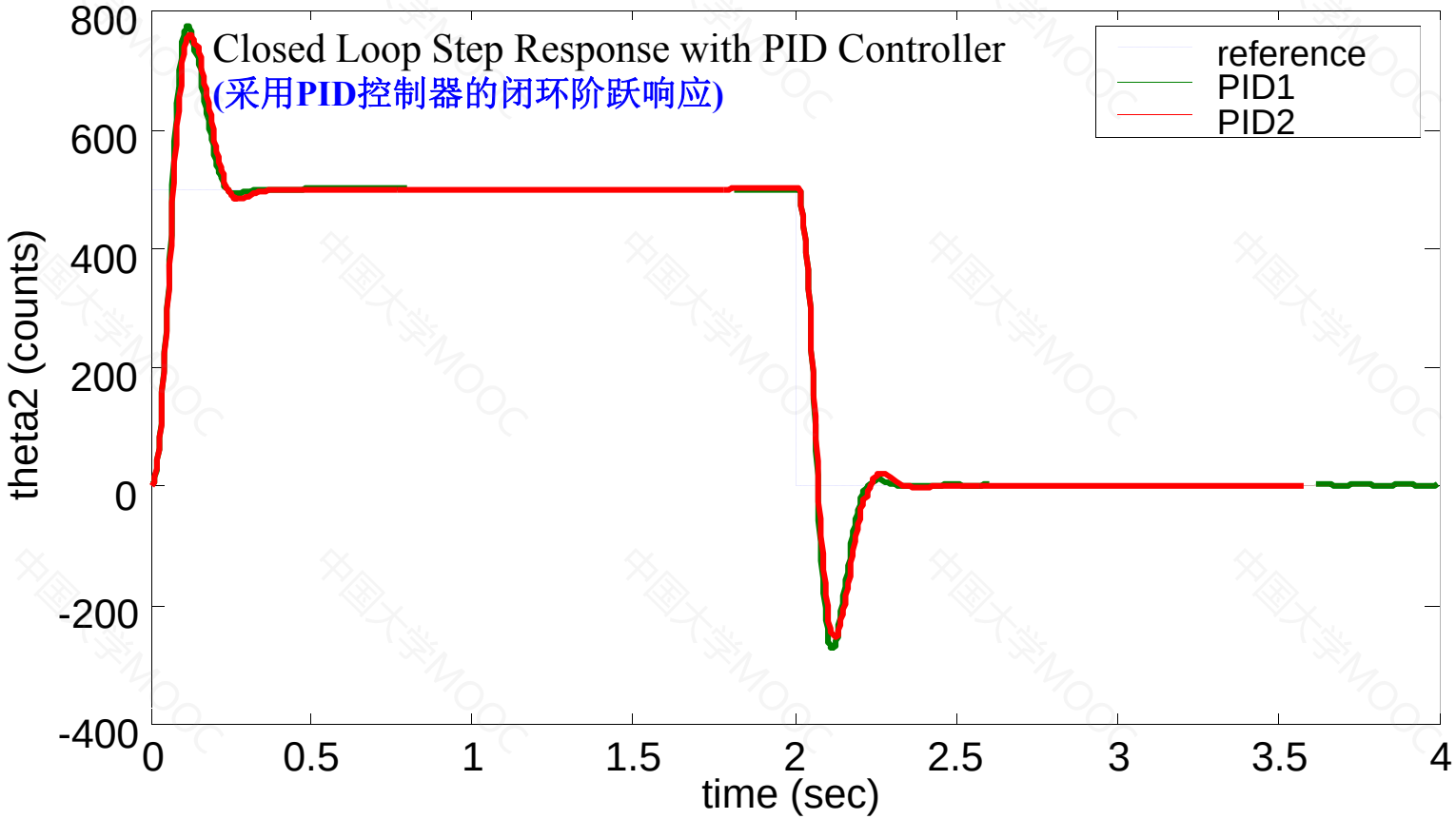
Actual CLCP (实际闭环特征多项式)：

$$A_{CL}(s) = \underbrace{(s^2 + a_{p1}s)}_{D_p}\underbrace{(s^2 + a_{c1}s)}_{D_c} + \underbrace{b_{p0}}_{N_p}(b_{c2}s^2 + b_{c1}s + b_{c0})$$

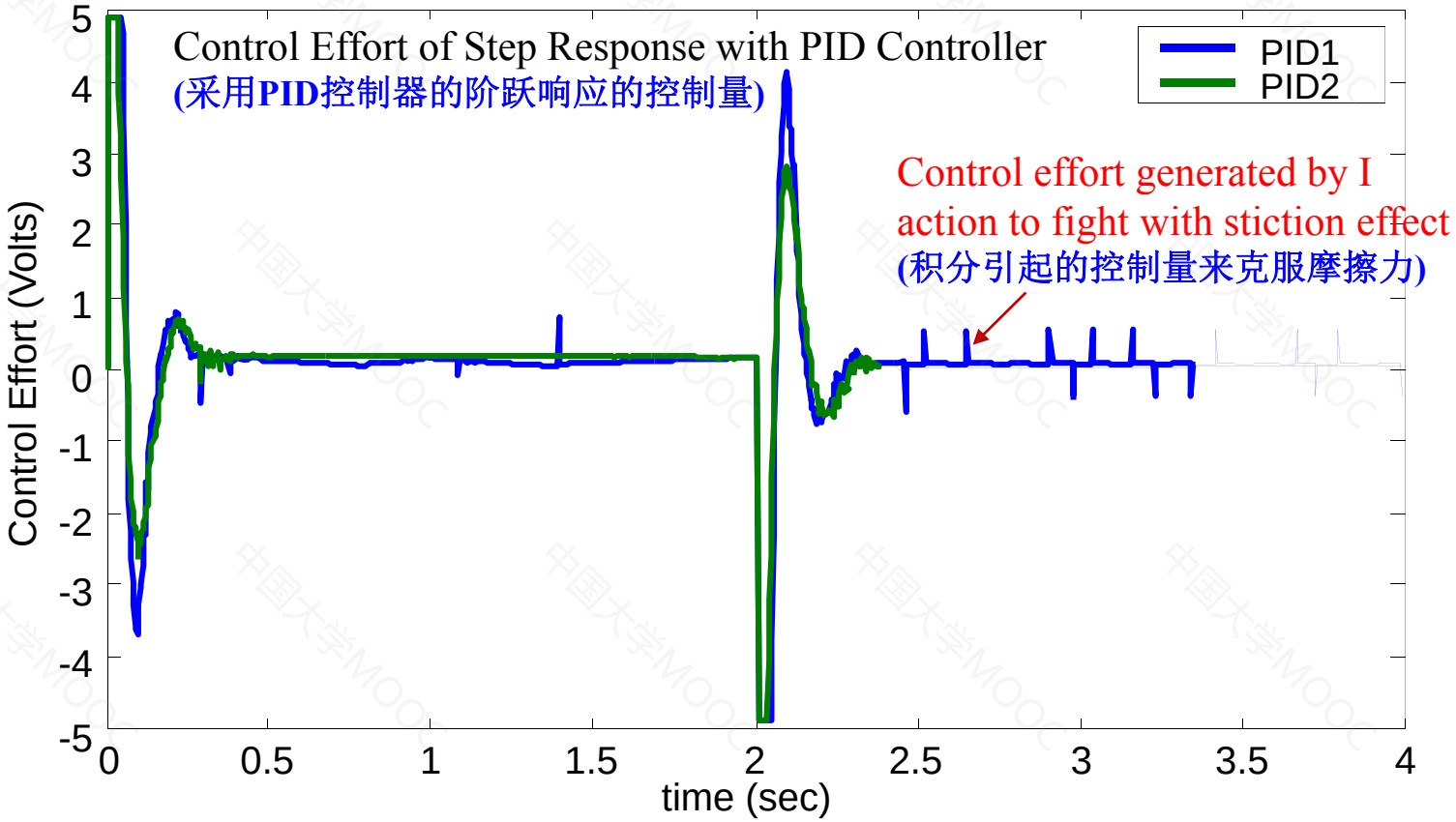
$$= s^4 + (a_{p1} + a_{c1})s^3 + (a_{p1}a_{c1} + b_{p0}b_{c2})s^2 + b_{p0}b_{c1}s + b_{p0}b_{c0}$$

$$\Rightarrow \begin{cases} a_{p1} + a_{c1} = 120 \\ a_{p1}a_{c1} + b_{p0}b_{c2} = 5600 \\ b_{p0}b_{c1} = 128000 \\ b_{p0}b_{c0} = 1280000 \end{cases} \Rightarrow \begin{cases} a_{c1} = 116.5 \\ b_{c2} = 8.33 \\ b_{c1} = 205 \\ b_{c0} = 2054 \end{cases}$$

PID Control Experimental Results



PID Control Experimental Results



PID Controller via Empirical Tuning

- **Ziegler-Nichols (Z-N) Tuning Rules (Z-N调整规则)**

This procedure is only valid for open loop stable plants and it is carried out through the following steps (这个过程只对开环稳定的系统有效，通过以下步骤进行)：

Step 1: Set the true plant under proportional control, increase the gain until the loop starts oscillating (对实际系统实施比例控制，增加增益直到回路开始振荡).

Step 2: Record the corresponding P-gain as the controller critical gain K_c and the oscillation period as T_c (将对应的P增益记为控制器临界增益K_c，振荡周期记为T_c).

Step 3: Adjust the controller parameters according to (按下表调整控制器参数)

	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.45K_C$	$0.54K_C/T_C$	
PID	$0.6K_C$	$1.2K_C/T_C$	$0.075K_CT_C$

Summary

- **PID Controller via Pole Placement**
  - Proper PID Controller Structure
  - Determine PID Gains via Pole Placement
    - τ_D is pre-determined
    - τ_D is automatically determined
  - Ziegler-Nichols (Z-N) Tuning Rules